

CAMPBREAK · PROBLEM STATEMENT

CampBreak problem statement

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	CBK-PRB-001	Version	0.4
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/campbreak		
Source	docs/01-problem.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-03	Amish Chadha	Estimated cost updated after Amish's R9 decision (CBK-DDR-003)
0.3	2026-10-03	Amish Chadha	TRL 3 with the decisions of CBK-DDR-001 and CBK-DDR-002 applied and results from CBK-CAL-001
0.2	2026-10-03	Amish Chadha	Populated to TRL 2 with concept media, components and first-order numbers
0.1	2026-09-30	Amish Chadha	Initial scaffold

CampBreak problem statement

In a dense camp, a shelter fire becomes a block fire within minutes. The people who are there in those minutes are residents, and they have almost nothing to fight it with.

■ The problem

Camp shelters are built from bamboo, tarpaulin and thatch, packed closer than planning standards allow, and cooked in with open flames. Residents in the January 2024 Camp 5 fire described flames spreading with the wind through thatch roofs, while hilly ground slowed vehicles and hydrants ran dry ([The New Humanitarian, 2024](#)). In informal settlements more widely, residents raise the alarm, evacuate, gather water and fight fires themselves, but with poor equipment and little training ([Engineering X](#)).

Parts of the answer exist separately. Lumkani's networked heat detectors raise the alarm across neighbouring homes and have been installed in South Africa, Kenya and Bangladesh ([Wikipedia](#)), and a 2026 study found rate-of-rise heat detection better suited than smoke alarms to homes that cook with open flames ([Afrin and Rush, 2026](#)). Camp agencies have deployed three-wheeler mobile firefighting units ([The New Humanitarian, 2024](#)). What is missing is an open kit that links the alarm to water at block level, is hand-powered, and can be built and repaired in the camp.

■ Users and context

User	Need	Context
Camp residents and block fire volunteers	Hear the alarm fast and get water onto a burning shelter within minutes	Night and day, narrow paths, slopes, no power at the shelter
Camp management and site planning agencies	A low-cost, repairable block kit that fits existing fire safety plans	Planned and unplanned camps, dry-season water shortages
Local fire services	Fires held small until engines or mobile units arrive	Long travel times and poor vehicle access
Community groups in urban informal settlements	A shared station they can own, maintain and train on	Dense shack neighbourhoods with communal standpipes

■ Operating environment

- Outdoor block station and indoor shelter alarms; ambient 5 to 45 °C (41 to 113 °F), monsoon rain, dust and high humidity.
- Unpaved, sloping and muddy paths, often under 1.5 m (5 ft) wide, with steps and drains to cross.
- Water from 200 L drums, jerrycans, tube wells or taps; supply may be low in the dry season.
- Cooking smoke and open flames inside or beside shelters, which rule out smoke-based detection.
- No mains power at the shelter; alarms run on batteries or small solar cells.

■ Constraints

- Value-engineering target of USD 1,600 for one block kit (24 alarms, one station and one hose cart), a hypothetical control target rather than a limit; off-the-shelf parts and workshop fabrication. Estimated cost of the constructable design: USD 1,374 (USD 226 under the target).
- Hand-powered pumping only; no fuel engine on the cart.
- Cart must be moved by two adults on the paths above.
- Alarm detection by rate of rise of temperature, not smoke, to avoid cooking false alarms.
- Open design: hardware under CERN-OHL-S-2.0, firmware under MIT.
- No claims of alarm or extinguisher certification; published as an engineering reference.

■ Out of scope

- Fighting fully developed block fires; that remains the fire service's job.
- Arson prevention, security and policing in camps.
- Shelter materials, fire-retardant treatments and site layout.
- Certified smoke or heat alarms and certified extinguishers.
- Motorised pumps and hydrant networks.

■ Prior work

Prior work	What it does	Gap for these users	Source
Lumkani networked heat detector	Rate-of-rise heat detector that alarms all homes within about 60 m and sends SMS alerts through a gateway	Raises the alarm but gives residents nothing to fight the fire with; the mesh network is described as patented	link
Stirrup pump	Portable hand pump used with a bucket of water to control small fires, used widely in civil defence	Needs constant bucket refilling and a crew; no stored hose or station at block level	link
Hand-pumped fire engines and hose carts (1700s to 1800s)	Manually pumped engines with hose, drawn by crews and fed by buckets or cisterns	Large crews and heavy apparatus; not sized for narrow camp paths or modern parts	link
Indian 5-gallon backpack firefighting pump	Backpack water tank with hand-operated pump used in wildland firefighting	Small water volume per trip and no alarm link	link
Camp mobile firefighting units (MFFU)	Three-wheeler firefighting vehicles deployed in the Cox's Bazar camps	Limited by terrain and vehicle access and by hydrant water in the dry season	link

■ Co-design

A camp management or shelter agency with an existing community fire volunteer programme, ideally working with a local fire service, so that the kit is tested against real drills, real paths and the water sources residents actually have.

■ Answers to the open questions

The scaffold's open questions were answered at TRL 2 and TRL 3 and decided under Amish's 2026-10-03 pre-approval (CBK-DDR-001):

- **Alarm relay scheme:** a published star scheme. Each alarm reports to the block station, which sounds its siren and broadcasts once so every alarm sounds. It is not a mesh, and it will be checked against Lumkani's patents before release (D1).
- **Kit size and coverage:** 24 alarms, one station and one cart per block; stations are sited so that no shelter is more than 70 m from one (D4, D9).
- **Flow and pressure:** about 20 L/min through a 6 mm jet reaching about 6.9 m, at about 1.2 bar at the pump, with a 4 bar relief valve (CBK-CAL-001, D7).
- **Theft and misuse:** the cart parks under the station roof in view of the block and the alarm box is a locked enclosure. Arson remains out of scope.
- **First drill:** the first candidate to approach is a camp management agency in the Cox's Bazar camps working with the Bangladesh Fire Service and Civil Defence; nothing is agreed yet (D10).

■ Safety

Safety: CampBreak is for small, early fires only. People leave first; nobody enters a burning shelter or stays when the fire spreads. Water is never used on burning cooking oil or live wiring. The pump and hose are pressurised (relief valve at 4 bar); the station holds a sealed lead-acid battery. The kit is an open engineering reference, not certified fire alarms or firefighting equipment.